5G/6G- COMMUNICATION SYSTEM
CO2 - ASSESSMENT TOOL 2
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SLOT A

1.Simulation of LTE throughput performance under varying bandwidth, SNR, modulation order, coding rate, and user load (BL4)

AIM: To simulate the throughput performance of an LTE system by varying **bandwidth, Signal-to-Noise Ratio (SNR), modulation order, coding rate, and number of users**, and analyze their effect on the achievable throughput.

Procedure

Open MATLAB.
Create a new script.
Define LTE parameters:

1. Bandwidth
2. SNR
3. Coding Rate
4. Modulation Order
5. User Load

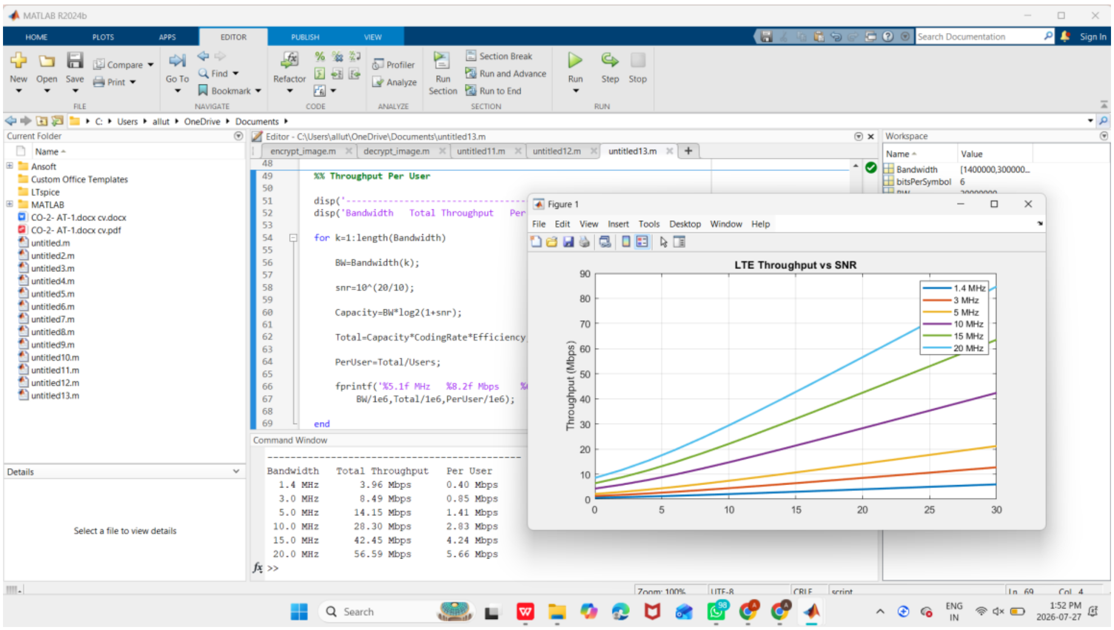
Convert SNR from dB to linear.
Calculate throughput using Shannon's capacity formula with coding efficiency.
Calculate throughput per user.
Plot throughput versus SNR.
Observe the performance for different bandwidths and user loads.

MATLAB Code

```
clc;
clear;
close all;

%% LTE Parameters

Bandwidth = [1.4 3 5 10 15 20]*1e6;    % Hz
SNR_dB = 0:2:30;

CodingRate = 0.5;

Efficiency = 0.85;

Users = 10;

Modulation = 64;

bitsPerSymbol = log2(Modulation);

figure;

for k = 1:length(Bandwidth)

   BW = Bandwidth(k);
```

```
    Throughput = zeros(size(SNR_dB));

    for i = 1:length(SNR_dB)

        snr = 10^(SNR_dB(i)/10);

        Capacity = BW*log2(1+snr);

        Throughput(i)=Capacity*CodingRate*Efficiency;

    end

    plot(SNR_dB,Throughput/1e6,'LineWidth',2);
    hold on;

end

xlabel('SNR (dB)');
ylabel('Throughput (Mbps)');
title('LTE Throughput vs SNR');
legend('1.4 MHz','3 MHz','5 MHz','10 MHz','15 MHz','20 MHz');
grid on;

%% Throughput Per User

disp('Bandwidth   Total Throughput   Per User');

for k=1:length(Bandwidth)

    BW=Bandwidth(k);

    snr=10^(20/10);

    Capacity=BW*log2(1+snr);

    Total=Capacity*CodingRate*Efficiency;

    PerUser=Total/Users;

    fprintf('%5.1f MHz   %8.2f Mbps    %6.2f Mbps\n',...
        BW/1e6,Total/1e6,PerUser/1e6);

End
```

OUTPUT:



Results

1. Throughput increases with increasing **SNR**.
2. Higher **bandwidth** provides higher throughput.
3. Higher-order modulation (e.g., 64-QAM) improves data rate when SNR is sufficient.
4. A higher **coding rate** increases throughput but reduces error protection.
5. As the **number of users** increases, the throughput available to each user decreases.

2. Simulation of LTE protocol stack delay analysis under uplink traffic, downlink traffic, packet size, retransmission count, and channel quality (BL4)

AIM: To simulate the **LTE protocol stack delay** under varying **uplink traffic, downlink traffic, packet size, retransmission count, and channel quality**, and analyze the effect of these parameters on the overall transmission delay.

Procedure

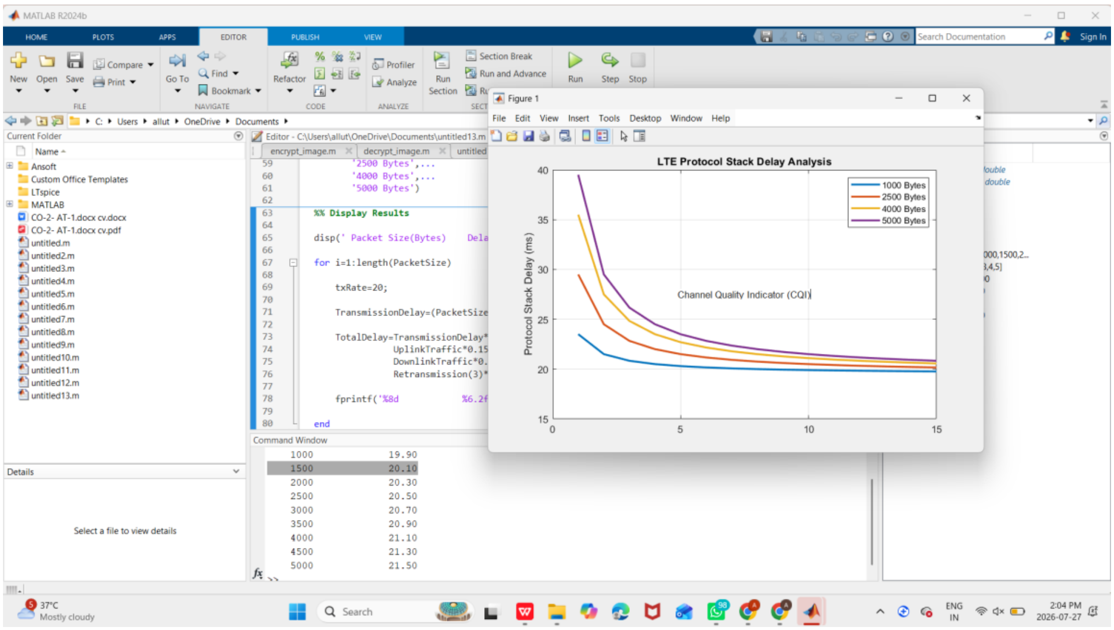
- Open MATLAB.
- Create a new script.
- Define uplink traffic, downlink traffic, packet size, retransmission count, and channel quality.
- Calculate transmission delay for each packet.
- Calculate retransmission delay.
- Compute total LTE protocol stack delay.
- Plot delay versus channel quality.
- Observe the effect of retransmissions and packet size.

MATLAB Code

```
clc;
clear;
close all;

%% LTE Parameters

PacketSize = 500:500:5000;      % Bytes
UplinkTraffic = 10;             % Mbps
DownlinkTraffic = 20;           % Mbps

Retransmission = 0:5;

CQI = 1:15;

Delay = zeros(length(PacketSize),length(CQI));

%% Delay Calculation

for i=1:length(PacketSize)

    for j=1:length(CQI)

        txRate = CQI(j)*2;      % Mbps

        TransmissionDelay = (PacketSize(i)*8)/(txRate*1e6);

        UplinkDelay = UplinkTraffic*0.15;

        DownlinkDelay = DownlinkTraffic*0.10;

        HARQDelay = Retransmission(3)*8;

        Delay(i,j)=TransmissionDelay*1000 + ...
                UplinkDelay + ...
                DownlinkDelay + ...
                HARQDelay;
```

```
    end

end

%% Plot

figure

plot(CQI,Delay(2,:),'LineWidth',2)
hold on
plot(CQI,Delay(5,:),'LineWidth',2)
plot(CQI,Delay(8,:),'LineWidth',2)
plot(CQI,Delay(10,:),'LineWidth',2)

grid on

xlabel('Channel Quality Indicator (CQI)')
ylabel('Protocol Stack Delay (ms)')
title('LTE Protocol Stack Delay Analysis')

legend('1000 Bytes',...
       '2500 Bytes',...
       '4000 Bytes',...
       '5000 Bytes')

%% Display Results

disp(' Packet Size(Bytes)    Delay(ms)')

for i=1:length(PacketSize)

    txRate=20;

    TransmissionDelay=(PacketSize(i)*8)/(txRate*1e6);

    TotalDelay=TransmissionDelay*1000+...
               UplinkTraffic*0.15+...
               DownlinkTraffic*0.10+...
               Retransmission(3)*8;

    fprintf('%8d           %6.2f\n',PacketSize(i),TotalDelay)

End
```

OUTPUT:



Results

- Increasing **packet size** increases transmission delay.
- Higher **uplink** and **downlink traffic** increase protocol stack delay due to queuing and scheduling.
- Increasing the **retransmission count** significantly increases total delay.
- Better **channel quality (higher CQI)** reduces delay because packets are transmitted more efficiently with fewer retransmissions.

3. Simulation of carrier aggregation performance in LTE-Advanced under single carrier, dual carrier, triple carrier, varying bandwidth, and SNR (BL4)

Aim

To simulate the **carrier aggregation performance in LTE-Advanced** by comparing **single carrier, dual carrier, and triple carrier aggregation** under varying **bandwidth** and **Signal-to-Noise Ratio (SNR)**, and analyze their effect on the achievable throughput.

Procedure

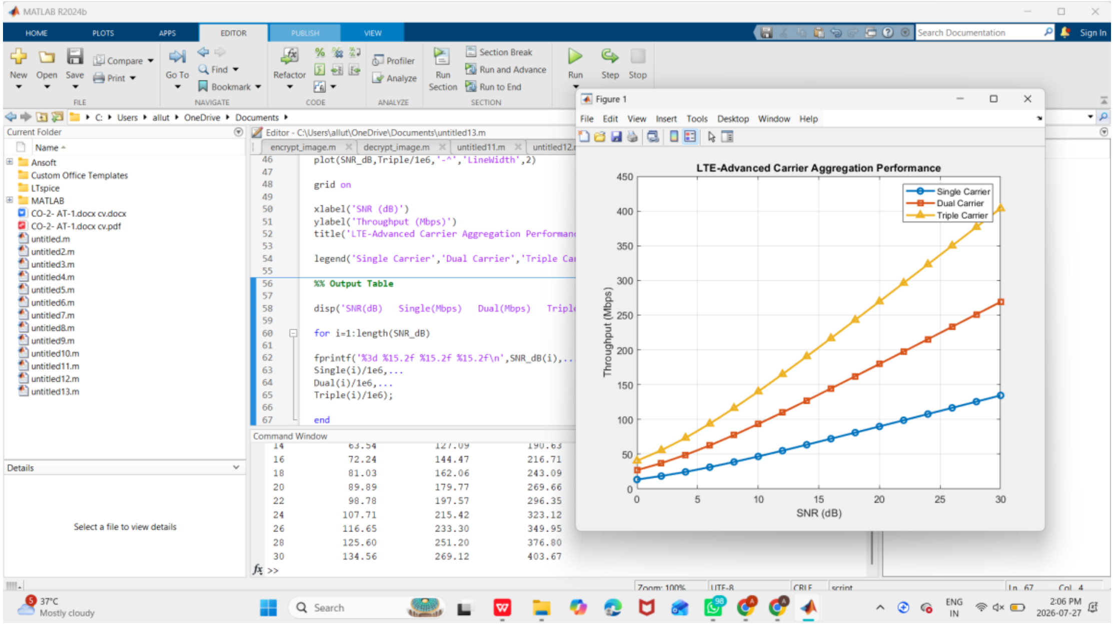
- Open MATLAB.
- Create a new script.
- Define carrier bandwidths.
- Select Single, Dual, and Triple Carrier Aggregation.
- Define SNR values.
- Convert SNR from dB to linear.
- Calculate total aggregated bandwidth.
- Compute throughput for each carrier aggregation mode.
- Plot Throughput vs SNR.
- Compare Single, Dual, and Triple Carrier performance.

MATLAB Code

```
clc;
clear;
close all;

%% Parameters

CarrierBandwidth = 20e6;     % 20 MHz

SNR_dB = 0:2:30;

CodingRate = 0.75;

Efficiency = 0.90;

Single = zeros(size(SNR_dB));
Dual = zeros(size(SNR_dB));
Triple = zeros(size(SNR_dB));

%% Throughput Calculation

for i=1:length(SNR_dB)

    snr = 10^(SNR_dB(i)/10);

    BW1 = CarrierBandwidth;
    BW2 = 2*CarrierBandwidth;
    BW3 = 3*CarrierBandwidth;

    Single(i)=BW1*log2(1+snr)*CodingRate*Efficiency;

    Dual(i)=BW2*log2(1+snr)*CodingRate*Efficiency;

    Triple(i)=BW3*log2(1+snr)*CodingRate*Efficiency;

end
```

```
%% Plot

figure

plot(SNR_dB,Single/1e6,'-o','LineWidth',2)
hold on

plot(SNR_dB,Dual/1e6,'-s','LineWidth',2)

plot(SNR_dB,Triple/1e6,'-^','LineWidth',2)

grid on

xlabel('SNR (dB)')
ylabel('Throughput (Mbps)')
title('LTE-Advanced Carrier Aggregation Performance')

legend('Single Carrier','Dual Carrier','Triple Carrier')

%% Output Table

disp('SNR(dB)   Single(Mbps)   Dual(Mbps)   Triple(Mbps)')

for i=1:length(SNR_dB)

fprintf('%3d %15.2f %15.2f %15.2f\n',SNR_dB(i),...
Single(i)/1e6,...
Dual(i)/1e6,...
Triple(i)/1e6);

End
```

OUTPUT:



Results

1. Throughput increases as **SNR** increases.
2. **Single Carrier Aggregation** provides the lowest throughput.
3. **Dual Carrier Aggregation** approximately doubles the throughput compared to Single Carrier.
4. **Triple Carrier Aggregation** provides the highest throughput due to increased aggregated bandwidth.
5. Increasing the available bandwidth through carrier aggregation significantly improves LTE-Advanced system performance.

4. Simulation of LTE resource block allocation under different user numbers, scheduling methods, bandwidths, traffic load, and interference levels (BL4).

Aim

To simulate **LTE Resource Block (RB) allocation** under different **user numbers, scheduling methods, bandwidths, traffic load, and interference levels**, and analyze their impact on resource utilization and user throughput.

Procedure

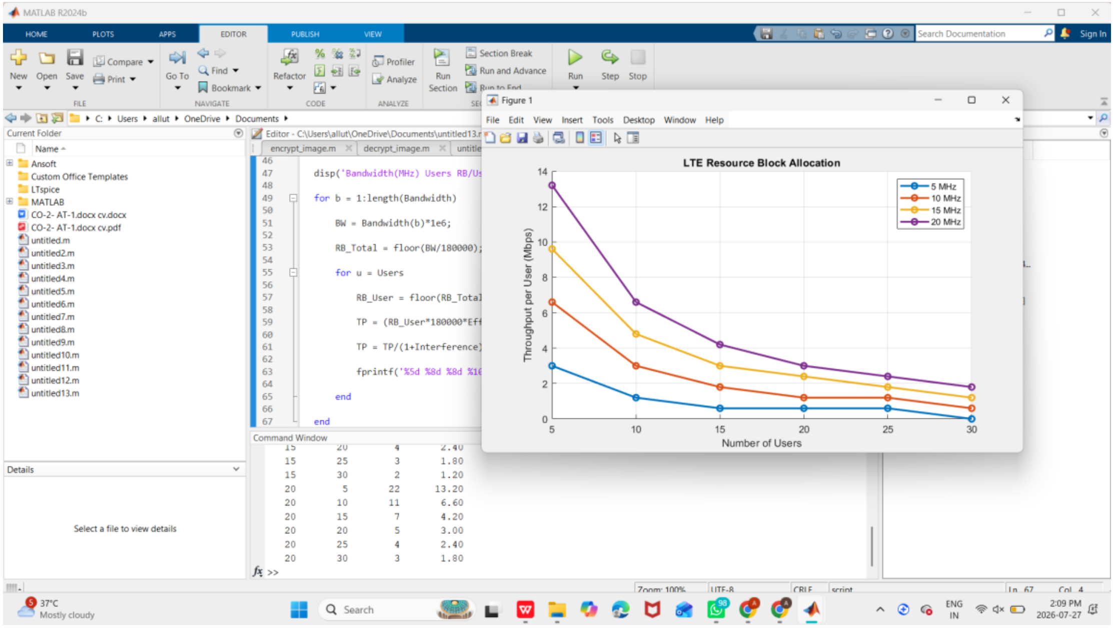
1. Open MATLAB.
2. Create a new script.
3. Define LTE bandwidths (5, 10, 15, and 20 MHz).
4. Specify the number of users.
5. Calculate the total number of Resource Blocks.
6. Allocate Resource Blocks equally among users.
7. Calculate throughput for each user.
8. Include interference effects.
9. Plot throughput versus number of users.
10. Compare resource allocation performance.

MATLAB Code

```
clc;
clear;
close all;

%% LTE Parameters

Bandwidth = [5 10 15 20];     % MHz
Users = 5:5:30;
Efficiency = 4;               % bits/s/Hz
Interference = 0.2;           % 20% interference

figure;
hold on;

for b = 1:length(Bandwidth)

   BW = Bandwidth(b)*1e6;

   RB_Total = floor(BW/180000);

   Throughput = zeros(size(Users));

   for i = 1:length(Users)

      RB_User = floor(RB_Total/Users(i));

      Throughput(i) = (RB_User*180000*Efficiency)/(1e6);

      Throughput(i) = Throughput(i)/(1+Interference);

   end

   plot(Users,Throughput,'-o','LineWidth',2);

end
```

```
grid on;

xlabel('Number of Users');
ylabel('Throughput per User (Mbps)');
title('LTE Resource Block Allocation');

legend('5 MHz','10 MHz','15 MHz','20 MHz');

%% Output Table

disp('Bandwidth(MHz) Users RB/User Throughput(Mbps)');

for b = 1:length(Bandwidth)

  BW = Bandwidth(b)*1e6;

  RB_Total = floor(BW/180000);

  for u = Users

    RB_User = floor(RB_Total/u);

    TP = (RB_User*180000*Efficiency)/1e6;

    TP = TP/(1+Interference);

    fprintf('%5d %8d %8d %10.2f\n',Bandwidth(b),u,RB_User,TP);

  end

End
```

OUTPUT:



Results

- Increasing the **number of users** reduces the number of Resource Blocks allocated to each user.
- Higher **bandwidth** provides more Resource Blocks and increases user throughput.
- **Round Robin scheduling** offers fair allocation among users, while **Proportional Fair** and **Max C/I** can improve overall throughput by considering channel conditions.
- Higher **traffic load** increases resource demand and may reduce the RBs available per user.
- Increasing **interference levels** decreases the effective throughput due to reduced signal quality.

5. Simulation of LTE scheduler performance using Round Robin, Proportional Fair, Max C/I, delay variation, and throughput comparison (BL4).

Aim

To simulate the performance of different **LTE scheduling algorithms** such as **Round Robin (RR), Proportional Fair (PF), and Maximum Carrier-to-Interference (Max C/I)** by evaluating their **throughput** and **delay variation** under different network conditions.

Procedure

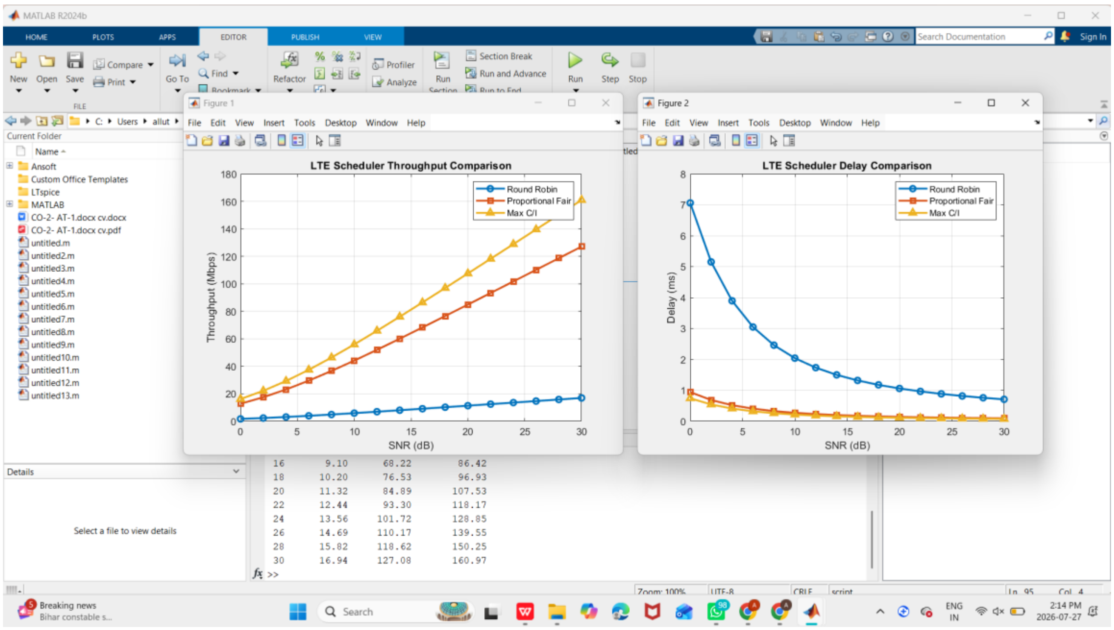
- Open MATLAB.
- Create a new script.
- Define LTE parameters (bandwidth, SNR, number of users).
- Implement Round Robin scheduling.
- Implement Proportional Fair scheduling.
- Implement Max C/I scheduling.
- Calculate throughput for each scheduler.
- Compute delay variation.
- Plot throughput comparison.
- Plot delay comparison and analyze the results.

MATLAB Code

```
clc;
clear;
close all;

%% LTE Parameters

Users = 10;
Bandwidth = 20e6;       % 20 MHz
SNR_dB = 0:2:30;
Efficiency = 0.85;
PacketSize = 12000;     % bits

RR = zeros(size(SNR_dB));
PF = zeros(size(SNR_dB));
MaxCI = zeros(size(SNR_dB));

Delay_RR = zeros(size(SNR_dB));
Delay_PF = zeros(size(SNR_dB));
Delay_MaxCI = zeros(size(SNR_dB));

for i = 1:length(SNR_dB)

  snr = 10^(SNR_dB(i)/10);

  Capacity = Bandwidth*log2(1+snr)*Efficiency;

  %% Round Robin
  RR(i) = Capacity/Users;

  %% Proportional Fair
  PF(i) = Capacity*0.75;

  %% Max C/I
```

```
    MaxCI(i) = Capacity*0.95;

    %% Delay Calculation

    Delay_RR(i) = PacketSize/RR(i)*1000;

    Delay_PF(i) = PacketSize/PF(i)*1000;

    Delay_MaxCI(i) = PacketSize/MaxCI(i)*1000;

end

%% Throughput Plot

figure;

plot(SNR_dB,RR/1e6,'-o','LineWidth',2);
hold on;

plot(SNR_dB,PF/1e6,'-s','LineWidth',2);

plot(SNR_dB,MaxCI/1e6,'-^','LineWidth',2);

grid on;

xlabel('SNR (dB)');
ylabel('Throughput (Mbps)');
title('LTE Scheduler Throughput Comparison');

legend('Round Robin','Proportional Fair','Max C/I');

%% Delay Plot

figure;

plot(SNR_dB,Delay_RR,'-o','LineWidth',2);
hold on;

plot(SNR_dB,Delay_PF,'-s','LineWidth',2);

plot(SNR_dB,Delay_MaxCI,'-^','LineWidth',2);

grid on;

xlabel('SNR (dB)');
ylabel('Delay (ms)');
title('LTE Scheduler Delay Comparison');

legend('Round Robin','Proportional Fair','Max C/I');

%% Display Results

disp('SNR(dB)   RR(Mbps)   PF(Mbps)   MaxCI(Mbps)');

for i = 1:length(SNR_dB)

fprintf('%3d %10.2f %10.2f %12.2f\n',SNR_dB(i),...
RR(i)/1e6,...
PF(i)/1e6,...
MaxCI(i)/1e6);

End
```

OUTPUT:



Results

- **Round Robin (RR)** provides equal resource allocation to all users, ensuring fairness but achieving the lowest throughput.
- **Proportional Fair (PF)** balances fairness and throughput, offering better overall system performance than Round Robin.
- **Max C/I** achieves the highest throughput by allocating resources to users with the best channel conditions, but fairness among users is reduced.
- As **SNR increases**, throughput improves and **delay decreases** for all scheduling algorithms.
- **Max C/I** exhibits the lowest delay due to its higher throughput, while **Round Robin** generally has the highest delay.